CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

Security Architecture and Threat Model

STRIDE Threat Modeling, OWASP Hardening, and Cryptographic Standards

DOCUMENT ID: 06_Security_Architecture_and_Threat_Model.pdf

Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. STRIDE Threat Model Analysis

A structured STRIDE threat assessment was executed across all trust boundaries:

Threat Category	Identified Attack Scenario	Implemented Security Control	Residual Risk
Spoofing	Attacker generates fake cashie	HMAC-SHA256 signature verifica	Negligible
Tampering	Shopper manipulates price in r	Server recalculates all prices	Zero
Repudiation	Cashier denies processing refu	Immutable audit_logs table rec	Zero
Info Disclosure	Credentials leaked in API resp	Password hashes & secrets stri	Negligible
Denial of Service	Inventory exhaustion via unpur	15-minute reservation TTL auto	Low
Elevation of Priv	Customer calls administrative	RolesGuard rejects unauthorize	Zero

2. Cryptographic & Data Protection Standards

- Password Hashing: SHA-512 / bcrypt with minimum 10 salt rounds.

- Token Architecture: Stateless 15-minute access tokens paired with revocable refresh tokens.

- SQL Injection Defense: All database operations execute through parameterized bindings.